

National University of Computer and Emerging Sciences, Lahore Campus



Course Name:	Electronic Devices and Circuits	Course Code:	EE1004
Program:	Electrical Engineering	Semester:	Fall2022
Duration:	180 Minutes	Total Marks:	100
Paper Date:	26-12-2022	Weight	50%
Section:	ALL	Page(s):	21
Exam Type:	Final		

Student : Name: _____ Roll No. _____ Section: _____

Instruction/Notes:

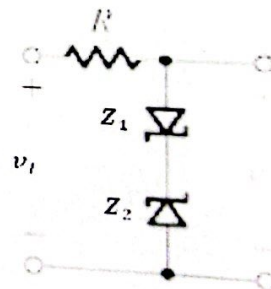
- Write your roll number on each page.
- Carefully read the paper and attempt each question only in the provided space.
- Show all calculations, direct answers are not acceptable. Your answers should be approximated up to three decimal places.
- Multiple solutions of same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

	Q1 (CLO#1)	Q2 (CLO#2)	Q3 (CLO#3)	Q4 (CLO#3)	Q5 (CLO#4)	Total
Total marks	20	20	20	20	20	100
Marks obtained						

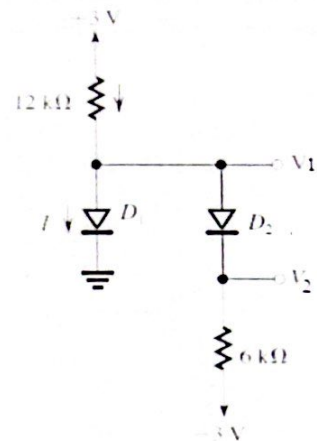
Question: 1 (CLO # 1)

20 Marks

- a) For the circuit shown below, assume the constant voltage drop model with $v_{Do} = 0.7V$ and $V_{z1} = 5.1V$, $V_{z2} = 6.8V$. Take $v_i = 20 \sin \omega t$
- Sketch and completely label the output waveform and the transfer characteristics.
 - Explain the working of two Zener diodes Z_1 and Z_2 in terms of biasing during one complete input cycle.



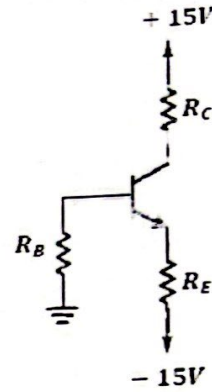
b) For the given circuit, Find the voltages V_1 , V_2 and current I . Use ideal diodes.



Question: 2 (CLO # 2)

20 Marks

It is required to design the circuit shown in figure below. A current of 2mA should be established in the emitter and a voltage of +6V should appear at the collector. The magnitude base voltage must be 2V. The transistor has a $\beta = 100$.



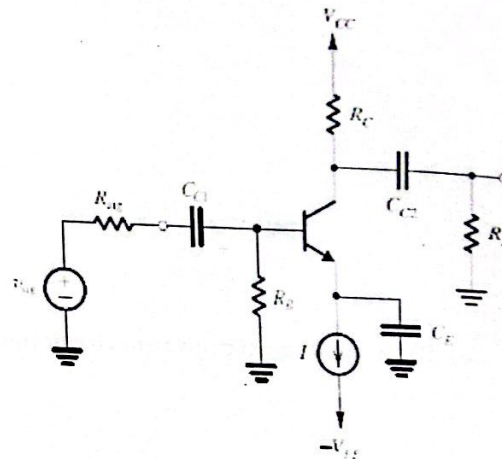
Question: 3 (CLO # 3)

20 Marks

Consider the following circuit with

$V_{CC} = V_{EE} = 15V$, $I = 2mA$, $R_B = 150 k\Omega$, $R_C = 5 k\Omega$,
 $\beta = 100$, $R_{sig} = 4 k\Omega$ and $R_L = 10 k\Omega$.

- a) Identify the configuration of the amplifier. Find all dc currents and voltages and confirm that BJT is in active mode.



b) Draw small signal equivalent circuit (include R_{sig} and R_L in your model) .

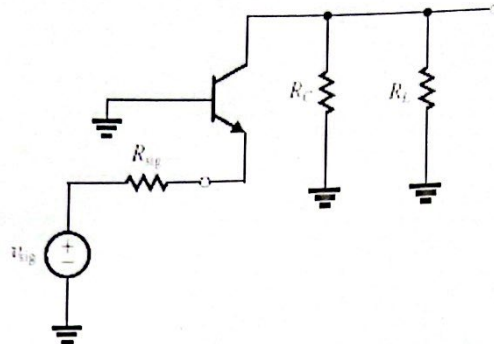
- c) Write the mathematical expressions and compute the value of input resistance, output resistance and the overall voltage gain G_V .

- d) The Designer of the amplifier decides to lower the bias current to half of its original value but maintain the overall voltage gain G_V to its original value, In order to do that, designer changes the value of R_C .
For the new design, determine the value of R_C and input resistance R_{in} .
Use new values of R_C and R_{in} to prove that voltage gain G_V remains unchanged.

Question: 4 (CLO # 3)

20 Marks

A Common Base amplifier, shown in figure below, is operating with $R_L = 5\text{ k}\Omega$, $R_C = 10\text{ k}\Omega$ and $R_{sig} = 100\text{ }\Omega$. Assume $\beta = 150$.



- a) Draw and completely label the small signal equivalent model of the amplifier.

b) At what current I_C should the transistor be biased for the input resistance R_{in} to equal that of the signal source?

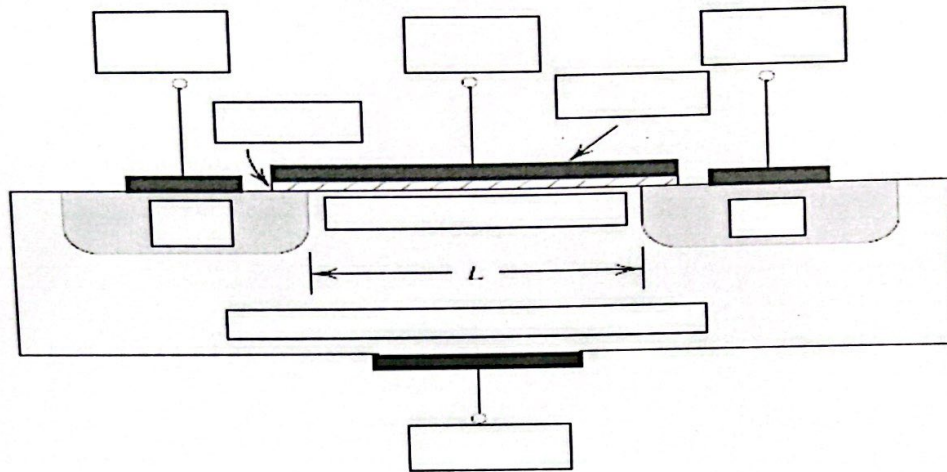
c) What is the resulting overall voltage gain?

- d) If the maximum signal amplitude of the voltage between base and emitter is limited to 10 mV, what are the corresponding amplitudes of v_{sig} and v_o ?

Question: 5 (CLO # 4)

20 Marks

- a) The physical structure of an N-channel MOSFET is shown below. Label the empty blocks.



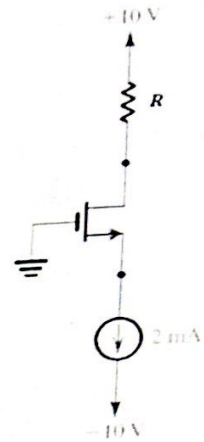
- b) Draw and label $i_D - v_{DS}$ characteristics of an N-channel enhancement MOSFET. (Show at least 4 curves for 4 different values of V_{GS})

- Identify the operating regions of NMOS on the characteristic curve. Explain how MOSFET works in each region.
- Give the operating conditions for V_{DS} and V_{GS} and expressions for drain current in each region.
- Comment on the shape of the channel in each region. Support your answer with figures.

c)

- i. Design the circuit shown in figure, transistor is characterized by

$$k'_n \frac{W}{L} = 1 \text{ mA/V}^2 \text{ and } V_t = 1 \text{ V}, V_D = 2 \text{ V}.$$



- ii. In the above figure, replace the current source with a resistor R_s and redraw the circuit. Design the circuit such that the current through R_s is same as the current source of 2mA.